

September Exam session Projects List

Objectives

The goal of the project is to tackle one of the proposed topics in the field of Computer Vision, developing non-trivial solutions. Students should explore the problem from an original perspective, applying methods that go beyond conventional solutions and demonstrating critical thinking and problem-solving skills.

Deadline

Students willing to deliver their project in the September exam session must fill [this Google Form](#) by September 6th.

Lines of conduct

- **Student groups:** The project can be carried out by a group, each consisting of a maximum of 3 people. Projects can also be completed by individual students, but we suggest to work in team.
- **Notebook format:** The project must be implemented using a notebook (e.g., Google Colab, Kaggle) or with an IDE (e.g., VSCode, PyCharm). The code should be optimized to support GPU usage and run without any error. The delivered code must follow the structure outlined below:
 - *Imports:* all the needed packages (for the notebook format)
 - *Globals:* useful variables on the whole code
 - *Utils:* code support functions
 - *Data:* everything related to data management
 - *Network:* code to structure the neural network
 - *Train:* part containing the training cycle elements
 - *Evaluation:* tests needed for the trained network

You can find a sample template at this [link](#). Try to maintain as much as possible this conceptual structure.

- **Deep learning framework:** All projects **MUST** be done in Python via the Pytorch framework.
- **Project assignment:** You are required to choose a project through [this Google Form](#). In this form, you will provide information about your team and the project, and include the link to the project's GitHub repository. In this repository you have to upload:
 - Code (or notebook) implementing the project
 - Dataset (or a link to it)
 - Project presentation
 - Detailed README to provide a quick overview of the project and instructions on how to run it
- **Project submission:** The project must be presented on one of the exam dates. It can be presented at a different time than the written exam. Both the written exam and the project **MUST** be completed within the academic year (i.e., between the June 2026 session and the March 2027 session).
- **Plagiarism:** Any attempt to plagiarize, whether by copying other students' work, directly replicating code from online resources, or submitting content highly retrieved from generative AI models, will be strictly penalized. This course values originality and personal effort; therefore, students must submit independently developed solutions. On the other hand, it is acceptable to consult external resources for inspiration or guidance. Violations will be subject to academic disciplinary actions, potentially including a failing grade for the project or further measures outlined by university policy.

Project 4: Emerging Interpretability in LeJEPA vs. Supervised Learning

Abstract: This project investigates the emergence of "zero-shot" semantic segmentation in Joint-Embedding Predictive Architectures (JEPAs) compared to traditional supervised models. While supervised training is driven by categorical labels, LeJEPA utilizes Sketched Isotropic Gaussian Regularization (SIGReg) to learn representations. We aim to evaluate whether the semantic structures emerging in LeJEPA's latent space (visible via PCA) align more naturally with the model's internal attention and gradients than those in supervised counterparts.

Dataset: ImageNet-1K, ImageNet-100, CIFAR-10, CIFAR-100.

Task: A comparative layer-by-layer analysis of feature localization and semantic consistency between LeJEPA-trained models and End-to-End Supervised models, utilizing both Vision Transformers (ViT) and Convolutional Neural Networks (CNNs).

Main objectives:

- *Model Training and Feature Extraction:* Deploy and contrast ViT and CNN architectures trained via LeJEPA and standard supervised learning. Extract latent representations across different layers for comparative probing.
- *Mapping XAI and Semantic Projections:* Implement a retrieval pipeline for explainable ai (XAI) saliency maps (using GMAR or Attention Rollout for ViTs and Grad-CAM for CNNs) and compute semantic masks through PCA projections of the embedding space.
- *Definition of a Semantic Alignment Score (SAS):* Design and implement a quantitative framework to measure the association between XAI-based saliency and PCA-based semantic components. The student should explore and justify the choice of metrics (e.g., spatial correlation, mutual information, or localized overlap) to define a robust score.
- *Qualitative and Comparative Analysis:* Conduct an in-depth visual study to assess if LeJEPA's "free" semantic segmentation provides a more interpretable representation of objects compared to the potentially biased focus of supervised models.

References:

- Balestrieri, R., & LeCun, Y. (2025). *LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics*.
- Jo, S., Jang, G., & Park, H. (2025). *GMAR: Gradient-Driven Multi-Head Attention Rollout for Vision Transformer Interpretability*.
- Selvaraju, R. R., et al. (2017). *Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization*.